Jintae Choi

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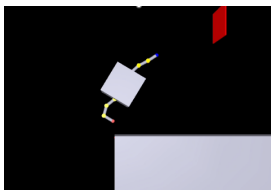
EDUCATION

Seoul National University Mar. 2025 – Present
Master's Student, Dept. of Aerospace Engineering
(항공우주공학과)
GPA: 3.65 / 4.3

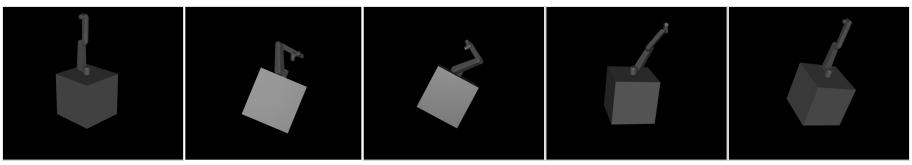
Seoul National University Mar. 2019 – Feb. 2025
B.S., Dept. of Mechanical & Aerospace Engineering
(기계항공공학부 우주항공공학전공)
GPA: 3.65 / 4.3, Major GPA: 3.76 / 4.3

RESEARCH INTERESTS

- **Space Robotics & Autonomous Systems**
  - **Learning based Nonholonomic Motion Planning:** Trajectory optimization for reorientation of Free-Floating Space Robots [1] (Falling cat dynamics).
  - **Physics-Informed Surrogate Modeling:** Enhancing trajectory optimization speed by replacing computationally expensive dynamic simulations with physics-constrained neural networks.
  - **Geometric Motion Planning:** Trajectory generation on manifolds for systems with nonholonomic constraints
  - **Reinforcement Learning for Space Robot:** Implementation of robust control policies for self-tossing maneuver.
- **Space Environment & System Engineering**
  - **Space Environment Simulation:** Operational experience with Thermal Vacuum Chambers (TVC) for satellite component qualification and environmental testing.
  - **Microgravity Experimental Systems:** Design and development of ground-based simulators to replicate microgravity conditions for space mechanism validation.



Jumping using RL



$t = 0.0s$ $t = 2.5s$ $t = 5.0s$ $t = 7.5s$ $t = 10.0s$

Reorientation using a generative model

[1] A nonholonomic system requiring consideration of dynamic coupling between the manipulator and the floating base.

EXPERIENCES

Space Mobility & Robotics Laboratory Seoul National University, Korea
Graduate Research Assistant Mar. 2025 – Present

- Conducting research on trajectory optimization for Free-Floating Space Robots subject to nonholonomic constraints.
- Managing and operating space environment testing facilities, including Thermal Vacuum Chambers.

Space Mobility & Robotics Laboratory Seoul National University, Korea
Research Intern Feb. 2024 – Feb. 2025

- Studied control algorithms for RPOD (Rendezvous, Proximity Operations, and Docking) and space robotics.
- Investigated ISAM (In-space Servicing, Assembly, and Manufacturing).
- Developed a testbed for ground simulation of space robots.

Bullnabi (Aerial Robotics Team) Seoul National University, Korea
Student Engineer Feb. 2023 – Oct. 2024

- Implemented trajectory generation algorithms for autonomous flight and landing.
- Designed and manufactured custom drone frames using 3D printing techniques.
- Validated flight control systems using SITL (Software-In-The-Loop) and HITL (Hardware-In-The-Loop) simulations.

Extreme Energy Laboratory Seoul National University, Korea
Research Intern Jul. 2023 – Sep. 2023

- Conducted thermal stability analysis of battery cells using Differential Scanning Calorimetry (DSC).
- Developed a thermal decomposition model in MATLAB based on the Arrhenius equation to analyze reaction kinetics.

PROJECTS

Trajectory Optimization for Free-Floating Space Robots Aug. 2025 – Present
Space Mobility & Robotics Lab

- Developed a hybrid motion planning framework specifically for nonholonomic base reorientation, integrating generative modeling with gradient-based optimization.
- Proposed a Physics-Informed Generative Model (PIGM) based on CVAE to generate valid warm-start trajectories satisfying angular momentum constraints, effectively mitigating local minima.
- Achieved machine-precision accuracy ($< 10^{-6}$ rad) and real-time performance by leveraging PyTorch-based differentiable simulation and validating in MuJoCo.
- Extended the optimization solver from Differentiable Physics (LBFGS) to iLQR to handle complex coupled dynamics and state constraints in high-dimensional configuration spaces.

Deep Reinforcement Learning for Space Manipulator Control Feb. 2024 – Jun. 2025
Undergraduate Researcher, Space Mobility & Robotics Lab

- Developed a high-fidelity 6-DOF floating-base simulation environment using MuJoCo to model complex contact dynamics in microgravity.
- Designed a TD3-based RL agent to perform self-tossing maneuvers, utilizing wall-pushing interactions to generate precise target momentum without propellant.
- Achieved robust control performance in departure velocity and direction, effectively handling contact uncertainties and impact disturbances.

Bio-inspired Tail-actuated Aerial Robot Sep. 2025 – Dec. 2025
Course Project: Biomimetic Robotics

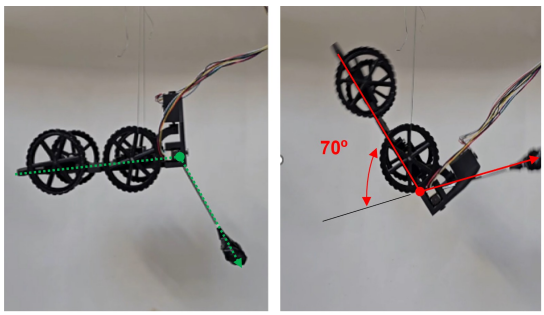
- Proposed and implemented a novel tail-based attitude control mechanism mimicking squirrel tail dynamics for aerial stability.
- Derived Lagrangian dynamics for the multi-body system to model the coupling between the tail and the fuselage.
- Built a functional prototype and validated the control logic through dynamic experiments.

Autonomous VTOL UAV Development (Team "Bullnabi") Dec. 2022 – Oct. 2024
Autonomous Landing Lead

- Spearheaded the autonomous landing team, developing a precision landing system using AprilTag vision and sensor fusion.
- Implemented Bézier curve-based trajectory generation algorithms to ensure smooth descent profiles.
- Validated flight logic using SITL/HITL simulations (PX4/Gazebo) prior to successful outdoor flight tests.

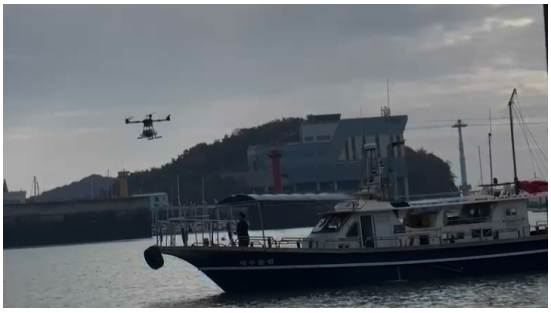
Other Projects & Achievements

- **Technology Transfer (Ship Landing):** Advanced the vision-based autonomous landing algorithm for maritime vessel environments and successfully transferred the technology to an industry partner.
- **Satellite Rendezvous Control:** Designed robust controllers (DOB+MPC) for precise relative motion under environmental disturbances (Course: Advanced Control Techniques).
- **Battery Thermal Runaway Modeling:** Modeled reaction kinetics using DSC data and nonlinear regression in MATLAB to analyze thermal stability.
- **Structural Failure Analysis:** Simulated submersible implosion scenarios using Ansys LS-Dyna (Course: Aircraft Structural Dynamics).



Initial condition Pitch control

Wheeled robot with a tail



Automation drone landing on a ship

PUBLICATIONS AND PRESENTATIONS

International Journal

[1] Jinhyuk Jang, **Jintae Choi**, and Hyeongjun Park, "Inertia-Free Task-Space Control of Free-Floating Space Manipulators for Orbital Manufacturing Applications," *Applied Sciences.* (**Major Revision**)

International Conference

[1] **Jintae Choi** and Hyeongjun Park, "Physics-Informed Generative Path Planning for Free-Floating Space Robot Orientation," *Submitted to IFAC 2026.* (**Under Review**)

Domestic Conference

[1] Jintae Choi, Hyeongjun Park, "우주 로봇팔을 이용한 비추력 방식 이동 강화학습 알고리즘," *2025년도 우주학술대회*, Yeosu, Korea (Jun. 2025) - Oral Presentation

PATENTS

[1] 박형준, 김민형, Patrick Ward, 최진태, "우주 물체 포획 장치", KR-Application No. 10-2025-0013919

[2] 박형준, 장진혁, 최진태, "오프셋 모터 기어트레인과 동심형 유체 로터리 조인트를 갖는 클리노스탯", KR-Application No. 10-2025-0151596

AWARDS AND HONORS

- 제 22회 한국로봇항공기 경연대회 우승, 산업통상자원부, Korea (Sep. 2024)

SKILLS

Programming:	Python, C++, MATLAB
Optimization & AI:	PyTorch, CasADi, Gurobi
Robotics & Sim:	ROS 2, MuJoCo, Gazebo, PX4
Embedded & HW:	NVIDIA Jetson, Arduino, 3D Printing
Design & CAD:	SolidWorks, Autodesk Inventor
Languages:	Korean (Native), English (OPIc IH)